

TalentLens AI

Intelligent Resume Parsing & AI-Driven Candidate Matching

Build what next India runs on

Team: [Your Team Name]

The Hiring Bottleneck

- HR teams spend up to 15 hours a week manually screening resumes.
- Traditional Keyword Matching is flawed-it misses highly qualified candidates who use synonyms.
- High volume of applications makes it impossible to give every candidate a fair evaluation.

Our Solution:

- An AI-powered matching engine that understands context, not just keywords.

JD Understanding & Candidate Evaluation

- Dynamic JD Parsing: We don't just extract text; we use AI to understand the intent of the JD.
- Smart Candidate Profiling: Resumes are parsed using NLP to build structured profiles.
- Beyond Keywords: If a JD asks for 'Machine Learning' and a resume says 'Trained Neural Networks',

our AI knows it's a match.

Ranking Methodology

Our algorithm scores candidates across 5 key dimensions:

- Technical Score (40%): Exact and partial skill overlap.
- Experience Score (25%): Textual similarity of past responsibilities + years of experience.
- Leadership Score (15%): Matching leadership expectations with past roles.
- Education Score (10%): Degree and field relevance.
- Soft Skills Score (10%): Inferred from project descriptions and achievements.

The LENS Assistant (Unique Feature)

Conversational AI for Recruiters:

- Interactive Chat: Recruiters can chat with our LENS Assistant (powered by Groq / Llama 3).
- Context-Aware: The AI knows all the candidates in the database.
- Use Cases:
 - 'Who is the best candidate for the Python Intern role?'
 - 'Draft a polite rejection email for Rahul.'
 - 'What are 3 technical interview questions I should ask Karthik?'

System Architecture

- Frontend: Next.js (React) dashboard with seamless Drag & Drop multi-file uploads.
- Backend: FastAPI (Python) asynchronous server ensuring non-blocking, high-speed API responses.
- AI Engine: Groq API running Llama-3.3-70b for lightning-fast semantic evaluation.
- Data Flow: Resume Upload -> Text Extraction -> Structuring -> Scoring Engine -> Dashboard.

Scalability & Workflow

- Bulk Processing: Recruiters can upload 10, 50, or 100 resumes at once.
- Asynchronous Queue: The backend processes resumes concurrently without freezing the UI.
- Fraud Risk Detection: Built-in AI flags suspicious resumes.
- Instant Comparisons: Select any two candidates for an AI-generated head-to-head comparison.

Results & Performance

- Time Saved: Reduces a 5-minute manual resume screen to < 3 seconds per candidate.
- Ultra-Low Latency: By utilizing Groq's LPU infrastructure, LLM evaluations happen instantly.
- Accuracy: Our custom 5-pillar scoring algorithm eliminates false positives.
- Zero Torch Dependency: Lightweight, math-based backend engine ensures high performance.

Technologies Used

- Frontend: Next.js, React, Tailwind CSS, Framer Motion, React Dropzone.
- Backend: Python, FastAPI, Pydantic, SQLAlchemy.
- AI & NLP: Groq (Llama-3.3-70b-versatile), pdfplumber.
- Deployment Setup: Ready for Vercel (Frontend) and Render (Backend).

Future Scope

- Automated Video Interview analysis.
- Integration with LinkedIn and GitHub for automatic portfolio verification.
- ATS (Applicant Tracking System) API integrations.

Live Demo Link: [Insert your Vercel Link here]

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THANK YOU